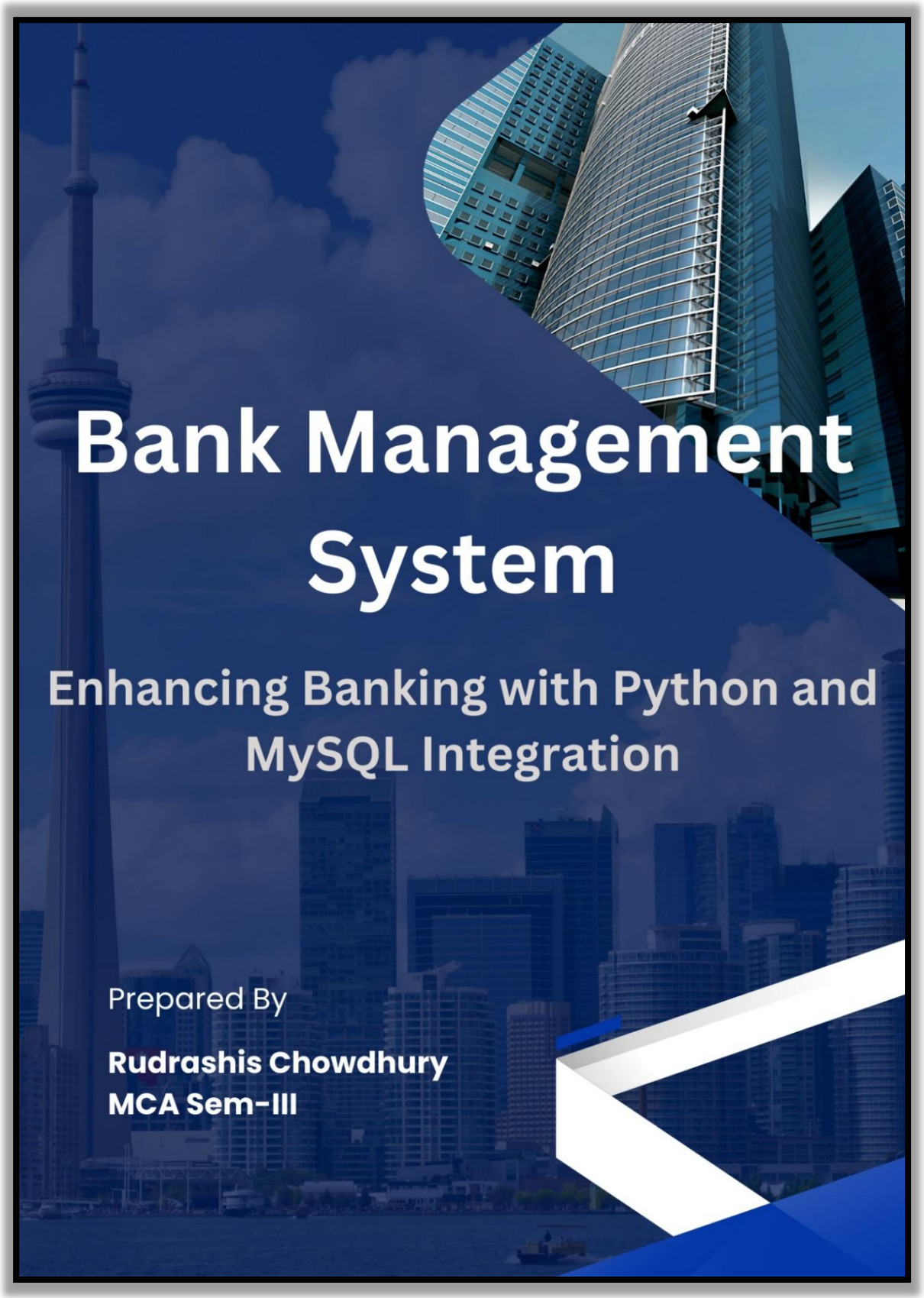

Bank Management System
Enhancing Banking with Python and MySQL Integration
Prepared By
Rudrashis Chowdhury
MCA Sem-III

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HARDWARES AND SOFTWARES

REQUIRED

HARDWARES

- **Desktop Computer/Laptop**
- **Windows 10**
- **Computers**
- **Server**
- **Database software (MySQL)**
- **Python development environment**

SOFTWARES

- **Python programming language**
- **MySQL database server**
- **Python MySQL Connector**
- **Operating system (Windows, Linux, or macOS)**



INTRODUCTION

Financial systems have always followed technological advancements. Banking, one of the oldest human institutions, went from ledger books to digital systems over the last century. This was driven by the need for efficiency and transparency. Initially, computers in banks were limited to standalone systems for basic operations. But with integrated systems and databases, banking changed, and we got real-time, customer centric services.

In the late 20th century, relational database management systems and programming languages changed how banks handled transactions. Python, a versatile and user-friendly language and MySQL, a powerful relational database system became the tools to automate banking services. Python and MySQL together provided a solid backend for banking applications and seamless data interaction. They bridged the gap between technical capabilities and user expectations.

Today we need systems that go beyond basic features. Bank Management System uses Python for its user-friendly interface and MySQL for robust data storage and management. This combination caters to both administrative and user needs from account management to real-time balance updates, loan tracking and user feedback system for seamless banking.

The project **Bank Management System** presented here represents a pivotal solution in the field of financial management, blending the sophistication of MySQL as the database backend with the dynamic capabilities of Python. This amalgamation results in a comprehensive system that caters to both administrative and user functionalities, streamlining processes related to account management, feedback collection, and loan tracking.

Evolution of Management Systems using Python and MySQL Connector

The evolution of management systems, especially in the banking sector, has witnessed a paradigm shift with the advent of powerful programming languages like Python and robust database management systems such as MySQL. Python, recognized for its simplicity, readability, and versatility, serves as the ideal frontend language for developing user-friendly interfaces. MySQL, on the other hand, provides a reliable and scalable database solution, ensuring the secure storage and retrieval of crucial banking information.

The integration of these technologies facilitates a seamless flow of data, enabling real-time interactions between users and the banking system. The dynamic connectivity offered by the MySQL Python Connector ensures efficient communication between the Python application and the MySQL database.

Purpose of Bank Management System

The Bank Management System project aims to address several key challenges in traditional banking operations. By leveraging Python's intuitive programming features and MySQL's robust data management capabilities, the system provides an agile and user-centric platform for both customers and administrators.

Objectives

The Overall objective of this research is to evaluate the implementation of Python and MySQL in bank management.

Specific objectives are:

- ✓ To develop a modular and scalable bank management system.
- ✓ To test the Python-MySQL integration in real-time.
- ✓ To measure user satisfaction and operational improvements.

Justification

This research is relevant because it follows the digital transformation in the banking industry. By using Python and MySQL, this research will provide insights in creating user friendly, secure and efficient systems. It will contribute to both theoretical and practical in financial technology.

Methodology

This research is a mixed method. The main investigation is to develop a prototype Bank Management System using Python and MySQL. Qualitative analysis will be done through user feedback on the system's interface and features. Quantitative metrics such as response time and data accuracy will be measured. The study will involve end-users and administrative personnel, using case studies to test the system in real-world.

Hypotheses

- Utilizing Python's user-friendly interface can improve the accessibility of banking operations for non-technical users.
- MySQL's robust database management system will enhance data security and operational reliability.
- A seamless connection between Python and MySQL via connector modules can provide real-time data synchronization, reducing processing time and errors.

User Functions

- ✓ **Add Account:** Users can easily open new accounts by providing essential details, streamlining the accountcreation process.
- ✓ **View Balance:** Account holders can check their account balance in real-time, fostering transparency and financial awareness.
- ✓ **Give Feedback:** The system facilitates a channel for users to provide feedback, ensuring continuous improvement based on user input.
- ✓ **View Transaction History:** Users have access to their transaction history, aiding financial record-keeping and reconciliation.

Admin Functions

- ✓ **Add New Accounts:** Administrators can add new accounts to the system, streamlining the process of onboarding new customers.
- ✓ **View Loan Details:** Admin have access to comprehensive loan details, enabling effectivemanagement and decision-making.
- ✓ **Update Loan Status:** The system empowers administrators to update the status of loans, trackrepayments and ensuring accurate record-keeping.
- ✓ **View Loan Defaulters:** Admin can identify and viewdetails of customers who have defaulted on loan payments for a specified period.
- ✓ **View User Feedback:** The system consolidates userfeedback, providing valuable insights into user experiences and concerns.



SOURCE CODE

```python
#BANK MANAGEMENT SYSTEM

#IMPORTING MYSQL IN PYTHON
#CONNECTING MYSQL WITH PYTHON
#IMPORT random FUNCTION FROM BUILT_IN FUNCTION
import mysql.connector as con
import random
print("\t---------------WELCOME TO BANK MANAGEMENT SYSTEM----------------")
print("")

#TO VIEW DATA OF THE USER
def view_data():
    dbo=con.connect(host="localhost" , user = "root" , password= "admin" ,database="bank_management_system")
    cu=dbo.cursor()
    number=int(input("Enter the account number"))
    query="select * from acct_holder where acct_no={}".format(number)
    cu.execute(query)
    for data in cu:
        print("*****************************************************************")
        print("Account number" , data[0])
        print("Name of account holder" , data[1])
        print("Phone number" , data[2])
        print("Email" , data[3])
        print("Address" , data[4])
        print("Intial balance " , data[5])
        print("Loan Taken" , data[6])
        print("*****************************************************************")
        n=int(input("""enter 1 for user menu :
                    enter 2 for exit: """))
        if n==1:
            user1()
        if n==2:
            return

```

```
36
37 #TO CHANGE THE NAME OF THE USER
38 def update_name():
39     dbo=con.connect(host='localhost' , user = 'root' , password= 'admin' ,database='bank_management_system')
40     cu=dbo.cursor()
41     holder=input('Enter the updated name')
42     number=int(input('Enter the account number'))
43     query='update acct_holder set acct_holder_name=('{}') where acct_no={}'.format(holder,number)
44     cu.execute(query)
45     dbo.commit()
46     print('----------------------------------------------------------------------')
47     n=int(input('''enter 1 for user menu :
48                 enter 2 for exit: '''))
49     print('Name successfully updated')
50     if n==1:
51             user1()
52     if n==2:
53             return
54
55 #TO CHANGE THE EMAIL OF THE USER
56 def update_email():
57     dbo=con.connect(host='localhost' , user = 'root' , password= 'admin' ,database='bank_management_system')
58     cu=dbo.cursor()
59     email=input('Enter the updated email')
60     number=int(input('Enter the account number'))
61     query='update acct_holder set email='{}' where acct_no={}'.format(email,number)
62     cu.execute(query)
63     dbo.commit()
64     print('Email  successfully updated')
65     print('-----------------------------------')
66     n=int(input('''enter 1 for user menu :
67                   enter 2 for exit: '''))
68     if n==1:
69         user1()
70     if n==2:
71         return
72
```

```
72
73 #TO CHANGE THE PHONE NUMBER OF THE USER
74 def update_phone_number():
75     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
76     cu=dbo.cursor()
77     phone=input("Enter the updated phone number")
78     number=int(input("Enter the account number"))
79     query="update acct_holder set phone_number={} where acct_no={}".format(phone,number)
80     cu.execute(query)
81     dbo.commit()
82     print("Phone number  successfully updated")
83     print("----------------------------------------")
84     n=int(input("""enter 1 for user menu :
85                    enter 2 for exit: """))
86     if n==1:
87         user1()
88     if n==2:
89         return
90
91 #TO CHANGE THE ADDRESS OF THE USER
92 def update_address():
93     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
94     cu=dbo.cursor()
95     address=input("Enter the updated address")
96     number=int(input("Enter the account number"))
97     query="update acct_holder set address='{}' where acct_no={}".format(address,number)
98     cu.execute(query)
99     dbo.commit()
100    print("Address was updated successfully")
101    print("-------------------------------------")
102    n=int(input("""enter 1 for user menu :
103                   enter 2 for exit: """))
104    if n==1:
105        user1()
106    if n==2:
107        return
108
```

```
108
109 #TO GIVE FEEDBACK TO THE BANK
110 def give_feedback():
111     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
112     cu=dbo.cursor()
113     number=int(input("Enter your account number:"))
114     print("**********Enter your feedback**********")
115     feed=input("")
116     query="insert into feedback values('{}' , {})".format(feed,number)
117     cu.execute(query)
118     dbo.commit()
119     print("Thank you for your feedback")
120     print("----------------------------------")
121     n=int(input("""enter 1 for user menu :
122                    enter 2 for exit: """))
123     if n==1:
124         user1()
125     if n==2:
126         return
127
128 #TO VIEW THE LOAN STATUS OF THE USER BY ADMIN
129 def view_loan_status():
130     number=int(input("enter your account number"))
131     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
132     cu=dbo.cursor()
133     query="select * from loan_acct where acct_no={}".format(number)
134     cu.execute(query)
135     for data in cu:
136         print(cu)
137     number.commit()
138     print("----------------------------------------")
139     n=int(input("""enter 1 for admin menu :
140                    enter 2 for exit: """))
141     if n==1:
142         user1()
143     if n==2:
144         return
145
```

```
145
146 #TO ADD DATA OF A NEW USER
147 def add_data():
148     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
149     cu=dbo.cursor()
150     a=random.random()*100000000000
151     n=input("enter the name of account holder")
152     p=int(input("enter the phone number"))
153     e=input("enter the email of account holder")
154     ad=input("enter the address of account holder")
155     b=int(input("enter the balance"))
156     t=input("enter whether loan taken or not")
157     query="insert into acct_holder values({},'{}',{},'{}','{}',{},'{}')".format(a,n,p,e,ad,b,t)
158     cu.execute(query)
159     dbo.commit()
160     print("----------------------------------------")
161     print("Successfully new account was added")
162     print("----------------------------------------")
163
164
165 #TO SEE THE FEEDBACKS OF THE USERS
166 def view_feedback():
167     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
168     cu=dbo.cursor()
169     query="select * from feedback"
170     cu.execute(query)
171     for data in cu:
172         print("**********************************************************")
173         print("Account number " , data[0])
174         print("Feedback" , data[1])
175         print("**********************************************************")
176
177
```

```
177
178 #TO VIEW THE LOAN DETAILS OF THE USER
179 def view_loan_details():
180     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
181     cu=dbo.cursor()
182     query="select * from loan_acct"
183     cu.execute(query)
184     for data in cu:
185         print("*********************************************************************")
186         print("Account number " , data[0])
187         print("Account Holder Name ",  data[1])
188         print("Loan Taken" , data[2])
189         print("Type of Loan" , data[3])
190         print("Status of Loan" , data[4])
191         print("Number of months from which loan's interest not paid" , data[5])
192         print("*********************************************************************")
193
194
195
196 #TO CHANGE THE LOAN STATUS OF THE USER
197 def update_status_loan():
198     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
199     cu=dbo.cursor()
200     number=int(input("enter the account number"))
201     stat=input("the status of the loan cleared or pending")
202     query="update loan_acct set status_of_loan=('{}') where acct_no = {}"
203     cu.execute(query.format(stat,number))
204     dbo.commit()
205     print("--------------------------------")
206     print("Successfully updated loan status")
207     print("--------------------------------")
208
209
```

```
210
211 #TO CHECK THE LOAN DEFAULTERS
212 def status_loan_defaulters():
213     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
214     cu=dbo.cursor()
215     m=int(input("enter the number of months "))
216     query="select * from loan_acct where number_of_months_from_which_interest_is_not_paid>={}".format(m)
217     cu.execute(query)
218     for data in cu:
219         print("*****************************************************************")
220         print("Account number " , data[0])
221         print("Account Holder Name ",  data[1])
222         print("Loan Taken" , data[2])
223         print("Type of Loan" , data[3])
224         print("Status of Loan" , data[4])
225         print("Number of months from which loan's interest not paid" , data[5])
226         print("*****************************************************************")
227
228
229 #TO ADD NEW DATA FOR LOAN
230 def add_data_loan():
231     dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
232     cu=dbo.cursor()
233     a=int(input("enter the account number:"))
234     n=input("enter the name of account holder:")
235     p=int(input("enter amount of loan taken:"))
236     e=input("enter the type of loan:")
237     ad=input("enter the status of loan:")
238     b=int(input("enter the number of months from which loan's interest not paid:"))
239     query="insert into loan_acct values({},'{}',{},'{}','{}',{})".format(a,n,p,e,ad,b)
240     cu.execute(query)
241     dbo.commit()
242     print("Successfully added loan data")
243     print("-----------------------------------")
244
245
```

```
245
246
247 #ADMIN MENU
248 def admin():
249     while True:
250         print("***************WELCOME ADMIN TO BANK MANAGEMENT SYSTEM***************")
251         print("Enter 1 adding the data of new accounts:")
252         print("Enter 2 for viewing data of account holder's loan:")
253         print("Enter 3 for updating status of loan lend:")
254         print("Enter 4 for viewing detials of loan defaulters:")
255         print("Enter 5 for viewing feedback of users:")
256         print("Enter 6 adding the data of new accounts into loan accounts:")
257         print("Enter 7 for exiting:")
258         print("-----------------------------------------------------------------")
259         print("-----------------------------------------------------------------")
260         ch=int(input("Enter your choice"))
261
262         if ch==1:
263             add_data()
264         elif ch==2:
265             view_loan_details()
266         elif ch==3:
267             update_status_loan()
268         elif ch==4:
269             status_loan_defaulters()
270
271         elif ch==5:
272             view_feedback()
273         elif ch==6:
274             add_data_loan()
275         else:
276             break
```

```
277
278 #USER MENU
279 def user1():
280     print("------------------------------------------------------------------")
281     print("\t***************WELCOME USER TO BANK MANAGEMENT SYSTEM***************")
282     print("")
283     print("----------------------WELCOME USER MENU---------------------------")
284     print("Enter 1 for viewing your bank account data")
285     print("Enter 2 for updating the name ")
286     print("Enter 3 for updating the email ")
287     print("Enter 4 for updating the phone number ")
288     print("Enter 5 for updating the address")
289     print("Enter 6 for giving feedback")
290     print("Enter 7 for viewing the status of your loan")
291     print("------------------------------------------------------------------")
292     print("------------------------------------------------------------------")
293     ch=int(input("Enter your choice:"))
294
295     if ch==1:
296         view_data()
297     elif ch==2:
298         update_name()
299     elif ch==3:
300         update_email()
301     elif ch==4:
302         update_phone_number()
303     elif ch==5:
304         update_adress()
305     elif ch==6:
306         give_feedback()
307     elif ch==7:
308         view_loan_status()
309
```

```
309
310
311 #MAIN MENU
312 while True:
313     print("**************MAIN MENU**************")
314     print("                                    ")
315     print("Enter 1 to open for admin")
316     print("Enter 2 to open for user")
317     print("Enter 3 for exit")
318     print("------------------")
319     inp=int(input("Enter your choice"))
320
321
322
323     if inp==1:
324         print("Enter username")
325         user=input("")
326         print("Enter password")
327         pasw=input("")
328         print("---------------------")
329         dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
330         cu=dbo.cursor()
331         query="select * from admin_data"
332         cu.execute(query)
333         for data in cu:
334             if data[0]==user and data[1]==pasw:
335                 admin()
336
```

```
337
338     if inp==2:
339         print("Enter username")
340         user=input("")
341         print("Enter password")
342         pasw=input("")
343         print("--------------------")
344         dbo=con.connect(host="localhost" , user = "root" , password= "admin" , database="bank_management_system")
345         cu=dbo.cursor()
346         query="select * from user_data"
347         cu.execute(query)
348         for data in cu:
349             if data[0]==user and data[1]==pasw:
350                 user1()
351     if inp==3:
352         print("Thank you")
353         break
354
```



DATABASE

USE OF DATABASE AND SHOW ALL TABLE IN THE DATABASE:

```
mysql> use BANK_MANAGEMENT_SYSTEM;
Database changed
mysql> SHOW TABLES;
+-----------------------------------------+
| Tables_in_bank_management_system        |
+-----------------------------------------+
| acct_holder                             |
| admin_data                              |
| bank_accounts                           |
| feedback                                |
| loan_acct                               |
| transaction_history                     |
| user_data                               |
+-----------------------------------------+
7 rows in set (0.02 sec)
```

DATA IN ACCOUNT HOLDER TABLE:

OUTPUT

```
mysql> select*from acct_holder;
+-------------+-----------------+--------------+---------------------------+-----------------+---------+--------------------+
| acct_no     | acct_holder_name| phone_number | email                     | address         | balance | whether_loan_taken |
+-------------+-----------------+--------------+---------------------------+-----------------+---------+--------------------+
| 123456790528| Rohit           |   8527419904 | rohit1230gmail.com        | 123 Main St     |    1000 | No                 |
| 234567892992| Jane Smith      |   9876543488 | rocky12340gmail.com       | 456 Oak St      |    2000 | Yes                |
| 366275559424| Mohit           |   7008116736 | mohit85230gmail.com       | Gajapati Nagar  |    8000 | no                 |
| 393924640768| rakesh          |   7485961216 | jyoti14780gmail.com       | Bima nagar      |  100000 | Yes                |
| 530818269184| Ricky           |   8855223296 | rocky12340gmail.com       | Krishna nagar   |   50000 | No                 |
| 700400599040| Ashis           |   7896541184 | ashish12340gmail.com      | bima nagar      |    7000 | yes                |
| 714367172608| S.k             |   7532195840 | sk12340gmail.com          | Kamapali        |    7000 | Yes                |
| 741258952704| adi             |   7896321536 | gou1230gmail.com          | bima nagar      |   10000 | yes                |
| 753621598208| Aditya          |   8596741120 | adi660gmail.com           | amabapua        |   20000 | yes                |
| 952431470736| Snehasis        |   7880945600 | snehasis45640gmail.com    | Krishna vihar   |   10000 | NO                 |
+-------------+-----------------+--------------+---------------------------+-----------------+---------+--------------------+
10 rows in set (0.01 sec)
```

DESCRIBE ADMIN TABLE:

```
mysql> select*from admin_data;
+----------+-----------+
| username | password  |
+----------+-----------+
| admin1   | password1 |
| admin2   | password2 |
| admin3   | password3 |
| admin4   | password4 |
| admin5   | password5 |
+----------+-----------+
5 rows in set (0.00 sec)
```

DATA IN ADMIN TABLE :

OUTPUT

```
mysql> DESC admin_data;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| username | varchar(100)| NO   | PRI | NULL    |       |
| password | varchar(100)| NO   |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

DESCRIBE FEEDBACK TABLE:

```
mysql> DESC feedback;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| acct_no  | float(12,0)  | NO   | PRI | NULL    |       |
| feedback | varchar(3000)| NO   |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

SHOW FEEDBACK OF USER:

OUTPUT

```
mysql> select*from feedback;
+--------------+-----------------------------------------------+
| acct_no      | Feedback                                      |
+--------------+-----------------------------------------------+
| 123456790528 | It was very smooth to use the system created  |
| 234567892992 | It was good but a lot of scope of improvement |
| 366275559424 | Nice work very good experience                |
| 393924640768 | Need to improve                               |
| 530818269184 | Very good experience                          |
| 700400599040 | Very nice facilities experience               |
| 714367172608 | Thank you for your awesome service            |
+--------------+-----------------------------------------------+
7 rows in set (0.00 sec)
```

DESCRIBE LOAN DETAILS TABLE:

```
mysql> DESC loan_acct;
+-----------------------------------------+-------------+------+-----+---------+-------+
| Field                                   | Type        | Null | Key | Default | Extra |
+-----------------------------------------+-------------+------+-----+---------+-------+
| acct_no                                 | float(12,0) | NO   | PRI | NULL    |       |
| acct_holder_name                        | varchar(100)| NO   |     | NULL    |       |
| amount_of_loan_taken                    | float(12,0) | NO   |     | NULL    |       |
| type_of_loan                            | varchar(100)| NO   |     | NULL    |       |
| status_of_loan                          | varchar(10) | NO   |     | NULL    |       |
| number_of_months_from_which_interest_is_not_paid | float(12,0) | NO |  | NULL    |       |
+-----------------------------------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

DATA IN LOAN DETAILS TABLE

OUTPUT

```
mysql> select*from loan_acct;
+-------------+-----------------+---------------------+--------------+----------------+-----------------------------------------+
| acct_no     | acct_holder_name| amount_of_loan_taken| type_of_loan | status_of_loan | number_of_months_from_which_interest_is_not_paid |
+-------------+-----------------+---------------------+--------------+----------------+-----------------------------------------+
|   741258944 | goutam          |                1000 | personal_loan| Pending        |                                       5 |
|   753621004 | Aditya          |                1000 | Home loan    | cleared        |                                      10 |
| 123456790528| John Doe        |                   0 | Personal Loan| PENDING        |                                       0 |
| 234567892992| Jane Smith      |                5000 | Home Loan    | Pending        |                                       2 |
| 366275559424| ashish          |                8000 | Home loan    | Cleared        |                                       6 |
+-------------+-----------------+---------------------+--------------+----------------+-----------------------------------------+
5 rows in set (0.00 sec)
```

DESCRIBE TRANSACTION TABLE:

```
mysql> DESC transaction_history;
+------------------+---------------+------+-----+-------------------+-----------------------------+
| Field            | Type          | Null | Key | Default           | Extra                       |
+------------------+---------------+------+-----+-------------------+-----------------------------+
| transaction_id   | int           | NO   | PRI | NULL              | auto_increment              |
| account_number   | int           | YES  | MUL | NULL              |                             |
| transaction_type | varchar(10)   | YES  |     | NULL              |                             |
| amount           | decimal(10,2) | YES  |     | NULL              |                             |
| transaction_date | timestamp     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED           |
+------------------+---------------+------+-----+-------------------+-----------------------------+
5 rows in set (0.00 sec)
```

SHOW TRANSACTION HISTORY OF THE USERS:




```
mysql> select*from transaction_history;
+----------------+----------------+------------------+-----------+---------------------+
| transaction_id | account_number | transaction_type | amount    | transaction_date    |
+----------------+----------------+------------------+-----------+---------------------+
|              1 |       40208143 | deposit          |   2000.00 | 2023-12-20 12:02:02 |
|              2 |       40208143 | withdraw         |   5000.00 | 2023-12-20 12:05:22 |
|              3 |       40208143 | deposit          | 400000.00 | 2023-12-20 12:06:12 |
|              4 |       40208143 | withdraw         | 400000.00 | 2023-12-20 12:10:21 |
|              5 |       40208143 | withdraw         |   1000.00 | 2023-12-20 12:15:52 |
+----------------+----------------+------------------+-----------+---------------------+
5 rows in set (0.00 sec)
```

DESCRIBE USER_DATA:

```
mysql> DESC user_data;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| username | varchar(100) | NO   | PRI | NULL    |       |
| password | varchar(100) | NO   |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

DATA IN USER TABLE:

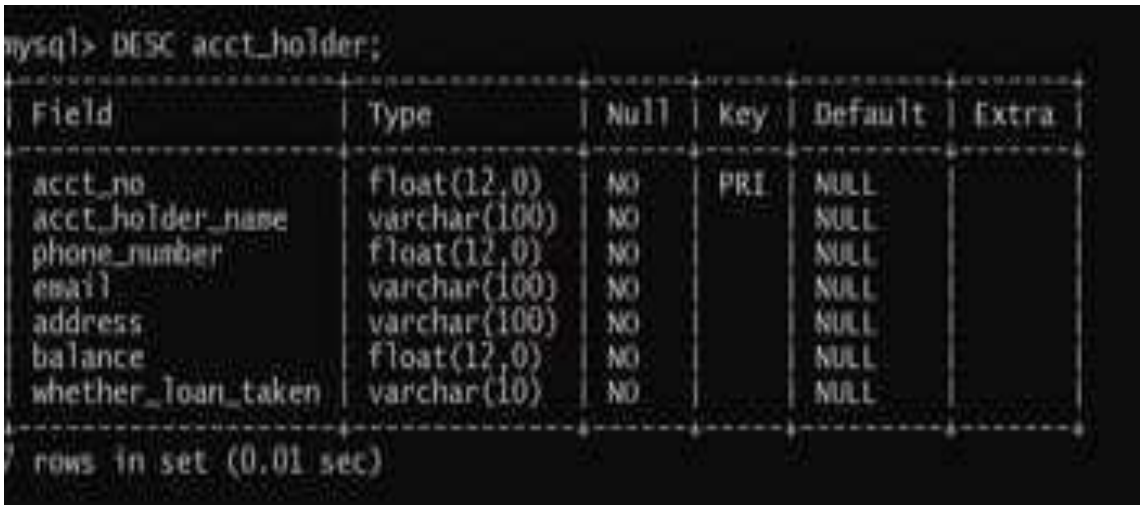
OUTPUT



```
mysql> select*from  user_data;
+----------+-----------+
| username | password  |
+----------+-----------+
| user1    | password1 |
| user2    | password2 |
| user3    | password3 |
| user4    | password4 |
| user5    | password5 |
+----------+-----------+
5 rows in set (0.00 sec)
```

ACCOUNT HOLDER DATA:

OUTPUT



```
mysql> DESC acct_holder;
+-------------------+-------------+------+-----+---------+-------+
| Field             | Type        | Null | Key | Default | Extra |
+-------------------+-------------+------+-----+---------+-------+
| acct_no           | float(12,0) | NO   | PRI | NULL    |       |
| acct_holder_name  | varchar(100)| NO   |     | NULL    |       |
| phone_number      | float(12,0) | NO   |     | NULL    |       |
| email             | varchar(100)| NO   |     | NULL    |       |
| address           | varchar(100)| NO   |     | NULL    |       |
| balance           | float(12,0) | NO   |     | NULL    |       |
| whether_loan_taken| varchar(10) | NO   |     | NULL    |       |
+-------------------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)
```

MAIN MENU:

```
----------WELCOME TO BANK MANAGEMENT SYSTEM----------

**********MAIN MENU**********

Enter 1 to open for admin
Enter 2 to open for user
Enter 3 for exit
------------------
Enter your choice1
Enter username
admin1
Enter password
password1
```

ADMIN MENU:

```
**********WELCOME ADMIN TO BANK MANAGEMENT SYSTEM**********
Enter 1 adding the data of new accounts:
Enter 2 for viewing data of account holder's loan:
Enter 3 for updating status of loan lend:
Enter 4 for viewing detials of loan defaulters:
Enter 5 for viewing feedback of users:
Enter 6 adding the data of new accounts into loan accounts:
Enter 7 for exiting:
--------------------------------------------------------------------------
```

ADDING THE DATA OF NEW ACCOUNT:

```
**********WELCOME ADMIN TO BANK MANAGEMENT SYSTEM**********
Enter 1 adding the data of new accounts:
Enter 2 for viewing data of account holder's loan:
Enter 3 for updating status of loan lend:
Enter 4 for viewing detials of loan defaulters:
Enter 5 for viewing feedback of users:
Enter 6 adding the data of new accounts into loan accounts:
Enter 7 for exiting:
--------------------------------------------------------------------------
--------------------------------------------------------------------------
Enter your choice1
enter the name of account holderS.k
enter the phone number7532195861
enter the email of account holderski1234@gmail.com
enter the address of account holderKamapali
enter the balance7000
enter whether loan taken or notYes
----------------------------------------
Successfully new account was added
```

FOR VIEWING DATA OF ACCOUNT HOLDERS LOAN:

```
**********WELCOME ADMIN TO BANK MANAGEMENT SYSTEM**********
Enter 1 adding the data of new accounts:
Enter 2 for viewing data of account holder's loan:
Enter 3 for updating status of loan lend:
Enter 4 for viewing detials of loan defaulters:
Enter 5 for viewing feedback of users:
Enter 6 adding the data of new accounts into loan accounts:
Enter 7 for exiting:
--------------------------------------------------------------------
--------------------------------------------------------------------
Enter your choice2
**********************************************************************
Account number  741258944.0
Account Holder Name  goutam
Loan Taken 1000.0
Type of Loan personal_loan
Status of Loan Pending
Number of months from which loan's interest not paid 5.0
**********************************************************************
**********************************************************************
Account number  7536216064.0
Account Holder Name  Aditya
Loan Taken 1000.0
Type of Loan Home loan
Status of Loan cleared
Number of months from which loan's interest not paid 10.0
**********************************************************************
**********************************************************************
Account number  123456790528.0
Account Holder Name  John Doe
Loan Taken 0.0
Type of Loan Personal Loan
Status of Loan cleared
Number of months from which loan's interest not paid 0.0
**********************************************************************
**********************************************************************
Account number  234567892992.0
Account Holder Name  Jane Smith
Loan Taken 5000.0
Type of Loan Home Loan
Status of Loan Pending
Number of months from which loan's interest not paid 2.0
**********************************************************************
```

FOR UPDATING STATUS OF LOAN LEND:

```
**********WELCOME ADMIN TO BANK MANAGEMENT SYSTEM**********
Enter 1 adding the data of new accounts:
Enter 2 for viewing data of account holder's loan:
Enter 3 for updating status of loan lend:
Enter 4 for viewing detials of loan defaulters:
Enter 5 for viewing feedback of users:
Enter 6 adding the data of new accounts into loan accounts:
Enter 7 for exiting:
--------------------------------------------------------------------
--------------------------------------------------------------------
Enter your choice3
enter the account number 123456790528
the status of the loan cleared or pendingPENDING
---------------------------------
Successfully updated loan status
---------------------------------
```

FOR VIEWING DETAILS OF LOAN DEFAULTERS:

```
**********WELCOME ADMIN TO BANK MANAGEMENT SYSTEM**********
Enter 1 adding the data of new accounts:
Enter 2 for viewing data of account holder's loan:
Enter 3 for updating status of loan lend:
Enter 4 for viewing detials of loan defaulters:
Enter 5 for viewing feedback of users:
Enter 6 adding the data of new accounts into loan accounts:
Enter 7 for exiting:
------------------------------------------------------------------
------------------------------------------------------------------
Enter your choice4
enter the number of months 5
**********************************************************************
Account number  741258944.0
Account Holder Name  goutam
Loan Taken 1000.0
Type of Loan personal_loan
Status of Loan Pending
Number of months from which loan's interest not paid 5.0
**********************************************************************
**********************************************************************
Account number  7536216064.0
Account Holder Name  Aditya
Loan Taken 1000.0
Type of Loan Home loan
Status of Loan cleared
Number of months from which loan's interest not paid 10.0
**********************************************************************
```

FOR VIEWING FEEDBACK OF USERS:

```
**********WELCOME ADMIN TO BANK MANAGEMENT SYSTEM**********
Enter 1 adding the data of new accounts:
Enter 2 for viewing data of account holder's loan:
Enter 3 for updating status of loan lend:
Enter 4 for viewing detials of loan defaulters:
Enter 5 for viewing feedback of users:
Enter 6 adding the data of new accounts into loan accounts:
Enter 7 for exiting:
------------------------------------------------------------------
------------------------------------------------------------------
Enter your choice5
**********************************************************************
Account number  123456790528.0
Feedback It was very smooth to use the system created
**********************************************************************
**********************************************************************
Account number  234567892992.0
Feedback It was good but a lot of scope of improvement
**********************************************************************
**********************************************************************
Account number  366275559424.0
Feedback Nice work very good experience
**********************************************************************
**********************************************************************
Account number  393924640768.0
Feedback Need to improve
**********************************************************************
**********************************************************************
Account number  530818269184.0
Feedback Very good experience
**********************************************************************
**********************************************************************
Account number  700400599040.0
Feedback Very nice facilities experience
**********************************************************************
```

ADDING THE DATA OF NEW ACCOUNTS INTO LOAN ACCOUNTS:

```
**************WELCOME ADMIN TO BANK MANAGEMENT SYSTEM**************
Enter 1 adding the data of new accounts:
Enter 2 for viewing data of account holder's loan:
Enter 3 for updating status of loan lend:
Enter 4 for viewing detials of loan defaulters:
Enter 5 for viewing feedback of users:
Enter 6 adding the data of new accounts into loan accounts:
Enter 7 for exiting:
-----------------------------------------------------------------------
-----------------------------------------------------------------------
Enter your choice6
enter the account number: 366275559424
enter the name of account holder:ashish
enter amount of loan taken:8000
enter the type of loan:Home loan
enter the status of loan:Cleared
enter the number of months from which loan's interest not paid:6
Successfully added loan data
```

USER MENU:

```
****************MAIN MENU***************
_____________________________________________
Enter 1 to open for admin
Enter 2 to open for user
Enter 3 for exit
--------------------
Enter your choice2
Enter username
user1
Enter password
password1
----------------------
```

FOR VIEWING YOUR BANK:

```
***************WELCOME USER TO BANK MANAGEMENT SYSTEM***************

------------------------WELCOME USER MENU-------------------------------
Enter 1 for viewing your bank account data
Enter 2 for updating the name
Enter 3 for updating the email
Enter 4 for updating the phone number
Enter 5 for updating the address
Enter 6 for giving feedback
Enter 7 for viewing the status of your loan
------------------------------------------------------------------------
------------------------------------------------------------------------
Enter your choice:1
Enter the account number 366275559424
************************************************************************
Account number 366275559424.0
Name of account holder ashish
Phone number 8523697664.0
Email omp7854@gmail.com
Address omm sahu
Intial balance  8000.0
Loan Taken no
************************************************************************
enter 1 for user menu :
                      enter 2 for exit: 1
------------------------------------------------------------------------
```

FOR UPDATING THE NAME:

```
***************WELCOME USER TO BANK MANAGEMENT SYSTEM***************

------------------------WELCOME USER MENU-------------------------------
Enter 1 for viewing your bank account data
Enter 2 for updating the name
Enter 3 for updating the email
Enter 4 for updating the phone number
Enter 5 for updating the address
Enter 6 for giving feedback
Enter 7 for viewing the status of your loan
------------------------------------------------------------------------
------------------------------------------------------------------------
Enter your choice:2
Enter the updated nameMohit
Enter the account number 366275559424
enter 1 for user menu :
                enter 2 for exit: 1
Name successfully updated
------------------------------------------------------------------------
```

FOR UPDATING THE MAIL:

```
***************WELCOME USER TO BANK MANAGEMENT SYSTEM***************

-------------------------WELCOME USER MENU-------------------------------
Enter 1 for viewing your bank account data
Enter 2 for updating the name
Enter 3 for updating the email
Enter 4 for updating the phone number
Enter 5 for updating the address
Enter 6 for giving feedback
Enter 7 for viewing the status of your loan
------------------------------------------------------------------------
------------------------------------------------------------------------
Enter your choice:3
Enter the updated emailmohit0523@gmail.com
Enter the account number 366275559424
Email  successfully updated
enter 1 for user menu :
                    enter 2 for exit: 1
------------------------------------------------------------------------
```

FOR UPDATING THE PHONE NUMBER:

```
***************WELCOME USER TO BANK MANAGEMENT SYSTEM***************

-------------------------WELCOME USER MENU-------------------------------
Enter 1 for viewing your bank account data
Enter 2 for updating the name
Enter 3 for updating the email
Enter 4 for updating the phone number
Enter 5 for updating the address
Enter 6 for giving feedback
Enter 7 for viewing the status of your loan
------------------------------------------------------------------------
------------------------------------------------------------------------
Enter your choice:4
Enter the updated phone number7008116936
Enter the account number 366275559424
Phone number  successfully updated
enter 1 for user menu :
                    enter 2 for exit: 1
------------------------------------------------------------------------
```

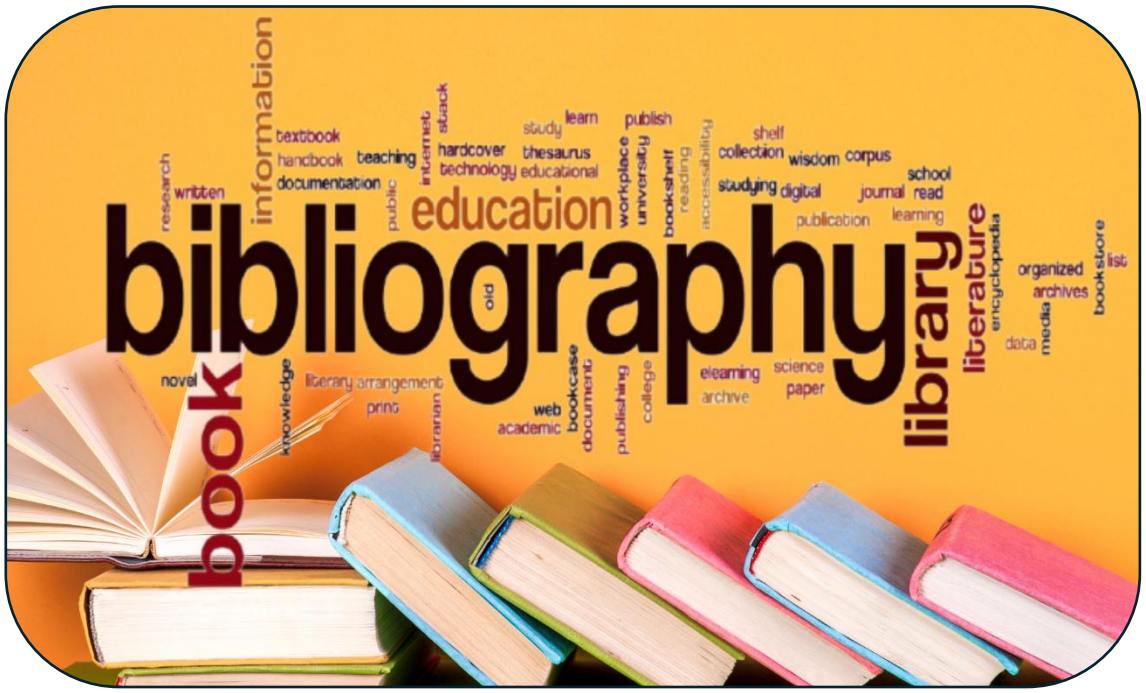
FOR UPDATING THE ADDRESS:

```
----------------------WELCOME USER MENU----------------------------
Enter 1 for viewing your bank account data
Enter 2 for updating the name
Enter 3 for updating the email
Enter 4 for updating the phone number
Enter 5 for updating the address
Enter 6 for giving feedback
Enter 7 for viewing the status of your loan
-------------------------------------------------------------------
-------------------------------------------------------------------
Enter your choice:5
Enter the updated addressGajapati Nagar
Enter the account number 366275559424
Address was updated successfully
enter 1 for user menu :
                  enter 2 for exit: 1
-------------------------------------------------------------------
```

FOR GIVING FEEDBACK:

```
***************WELCOME USER TO BANK MANAGEMENT SYSTEM***************

-----------------------WELCOME USER MENU----------------------------
Enter 1 for viewing your bank account data
Enter 2 for updating the name
Enter 3 for updating the email
Enter 4 for updating the phone number
Enter 5 for updating the address
Enter 6 for giving feedback
Enter 7 for viewing the status of your loan
-------------------------------------------------------------------
-------------------------------------------------------------------
Enter your choice:6
Enter your account number:714367139845
************Enter your feedback************
Thank you for your awesome service
Thank you for your feedback
```



1. **PYTHON**
2. **MY SQL**
3. **ARIHANT COMUTER SCIENCE WITH PYTHON**
4. **WIKIPEDIA**

Implementation:

- Planning: Define the project scope, objectives, and deliverables. Establish timelines and milestones. Determine the resources and team duties.
- Development: Use agile development methodology for iterative development and testing. Implement Python and MySQL according to the system design. Create a user-friendly GUI and ensure cross-platform compatibility.
- Testing: Perform unit, integration, and user acceptance testing. Verify functionality, performance, and security.
- Deployment: Deploy the system on an appropriate server infrastructure. Ensure correct configuration and data migration. Provide user training and documentation.
- Maintenance: Monitor system performance and user comments. Apply frequent updates and security fixes. Offer continued support and upgrades.
- Collaboration: Involve all stakeholders, including business users, the technical team, and management. Establish clear communication routes and provide regular progress reports.
- Risk Management: Identify potential risks and devise mitigating methods. Establish backup and disaster recovery plans.
- Continuous Improvement: Regularly collect comments and find opportunities for improvement. Implement improvements and modifications to optimize the system.

This implementation plan provides a structured approach for the Bank Management System's successful creation, deployment, and maintenance, ensuring its efficiency and long-term value.

Conclusion:

The Bank Management System is a comprehensive and user-friendly system that simplifies banking procedures and improves the overall user experience. The system uses Python and MySQL to provide extensive account administration, loan processing, feedback collection, and reporting features. Its scalable architecture and focus on security ensure the system's dependability and endurance.
We established a system that matches the changing needs of the banking industry by utilizing agile development processes, implementing rigorous testing procedures, and building a collaborative and risk-aware approach. The Bank Management System enables banks to improve operational efficiency, increase customer happiness, and maintain competitiveness in a continuously changing financial world.

Future Work:

The Bank Management System provides a solid foundation for further development and innovation. Future work in this area can focus on the following aspects:

- Mobile Integration: Creating a mobile application or responsive web design to expand system functionality to mobile devices, allowing for banking at any time and from any location.
- Artificial Intelligence and Machine Learning: Incorporating AI and machine learning algorithms to improve fraud detection, credit risk assessment, and personalized banking advice.
- Blockchain Integration: Exploring the use of blockchain technology to improve financial transaction security, transparency, and efficiency.
- Chatbot and Virtual Assistant Integration: Implementing chatbots and virtual assistants to deliver real-time customer service, automate tedious processes, and boost user engagement.
- Data Analytics and Business Intelligence: Creating advanced data analytics and business intelligence capabilities to analyse banking data, detect trends, and generate actionable insights for decision-making.

By leveraging these advancements, the Bank Management System can evolve into a comprehensive and future-proof solution that meets the ever-changing demands of the banking industry.

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We would want to take this opportunity to sincerely thank everyone who helped us reach this victorious end of our trip. Above all, we give thanks to the Almighty for His blessings, direction, and grace, which have been with us at every turn.

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